TASK22: Cloud App

AIM: To develop a cloud comparison application that displays and compares services and pricing models of AWS, GCP, and Azure

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design user interface with dropdown for cloud providers
4. Initialize cloud service data (AWS, GCP, Azure) in database module
5. Capture selected provider using JavaScript
6. Send request to server to fetch service details
7. Display compute, storage, database, and networking services
8. Accept user input for compute hours and storage usage
9. Send input data to server for cost calculation
10. Calculate cost based on pay-as-you-use pricing model
11. Display total cost dynamically on UI
12. Stop the program

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>
<html>
<head>
   <title>Cloud Comparison</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
```

```
    <h2>Cloud Service Comparison</h2>
    <select id="provider" onchange="loadData()">
        <option value="aws">AWS</option>
        <option value="gcp">GCP</option>
        <option value="azure">Azure</option>
    </select>
    <h3>Services</h3>
    <ul id="services"></ul>
    <h3>Pricing Calculator</h3>
    <input id="hours" placeholder="Compute Hours">
    <input id="storage" placeholder="Storage (GB)">
    <button onclick="calculate()">Calculate Cost</button>
    <h3 id="result"></h3>
</div>
<script src="script.js"></script>
</body>
</html>
```

STYLE.CSS:

```
body {
    font-family: Arial;
    background: #1e1e2f;
    color: white;
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
```

```
}
.container {
    text-align:center;
    width: 350px;
}
input, select, button {
    margin: 5px;
    padding: 8px;
}
li {
    background: #333;
    margin: 5px;
    padding: 5px;
}
```

SCRIPT.JS:

```
function loadData() {
    const provider = document.getElementById("provider").value;
    fetch(`/services/${provider}`)
    .then(res => res.json())
    .then(data => {
        let list = document.getElementById("services");
        list.innerHTML = "";
        Object.entries(data).forEach([key, value]) => {
            let li = document.createElement("li");
            li.innerText = `${key}: ${value}`;
            list.appendChild(li);
```

```
        });
    });
}
function calculate() {
    const provider = document.getElementById("provider").value;
    const hours = document.getElementById("hours").value;
    const storage = document.getElementById("storage").value;
    fetch("/cost", {
        method: "POST",
        headers: {"Content-Type":"application/json"},
        body: JSON.stringify({ provider, hours, storage })
    })
    .then(res => res.json())
    .then(data => {
        document.getElementById("result").innerText = "Cost: $" + data.cost;
    });
}
loadData();
```

SERVER.JS:

```
const express = require("express");
const db = require("./db");
const app = express();
app.use(express.json());
app.use(express.static(__dirname));
app.get("/services/:provider", (req, res) => {
    res.json(db.getServices(req.params.provider));
```

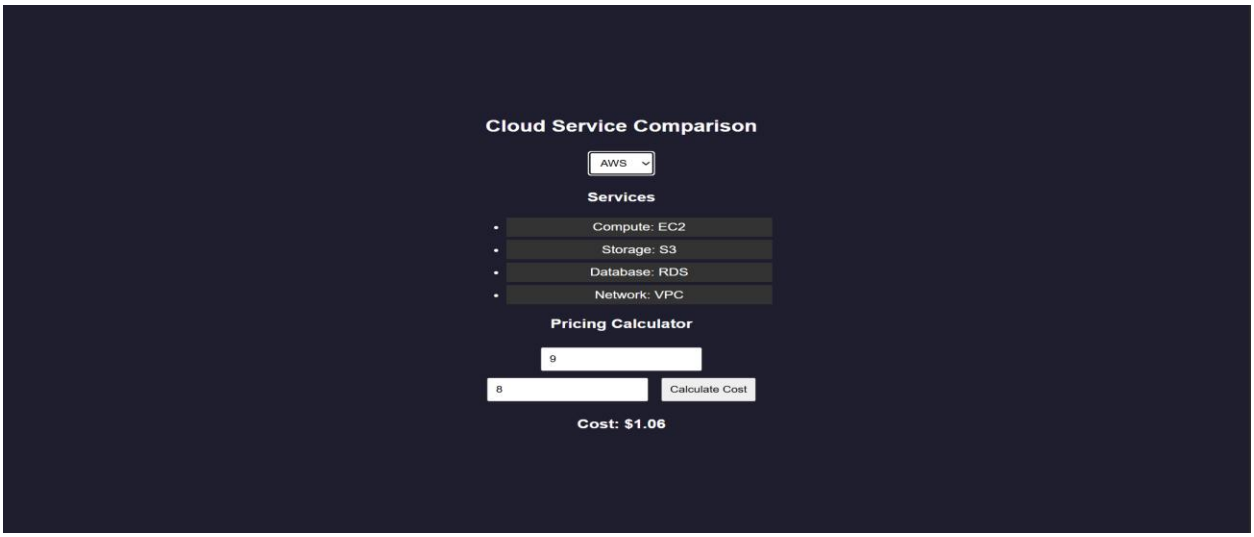
```
});
app.post("/cost", (req, res) => {
    const { provider, hours, storage } = req.body;
    const cost = db.calculate(provider, hours, storage);
    res.json({ cost });
});
app.listen(3000, () => {
    console.log("http://localhost:3000");
});
```

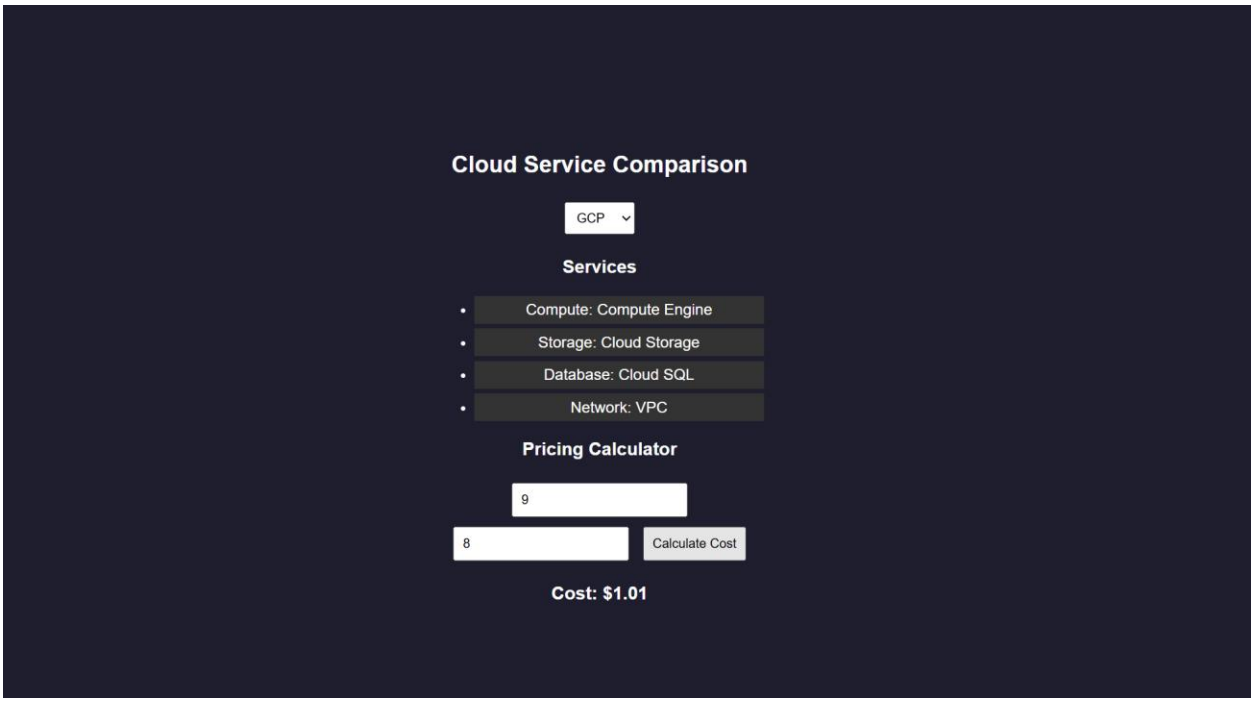
DB.JS:

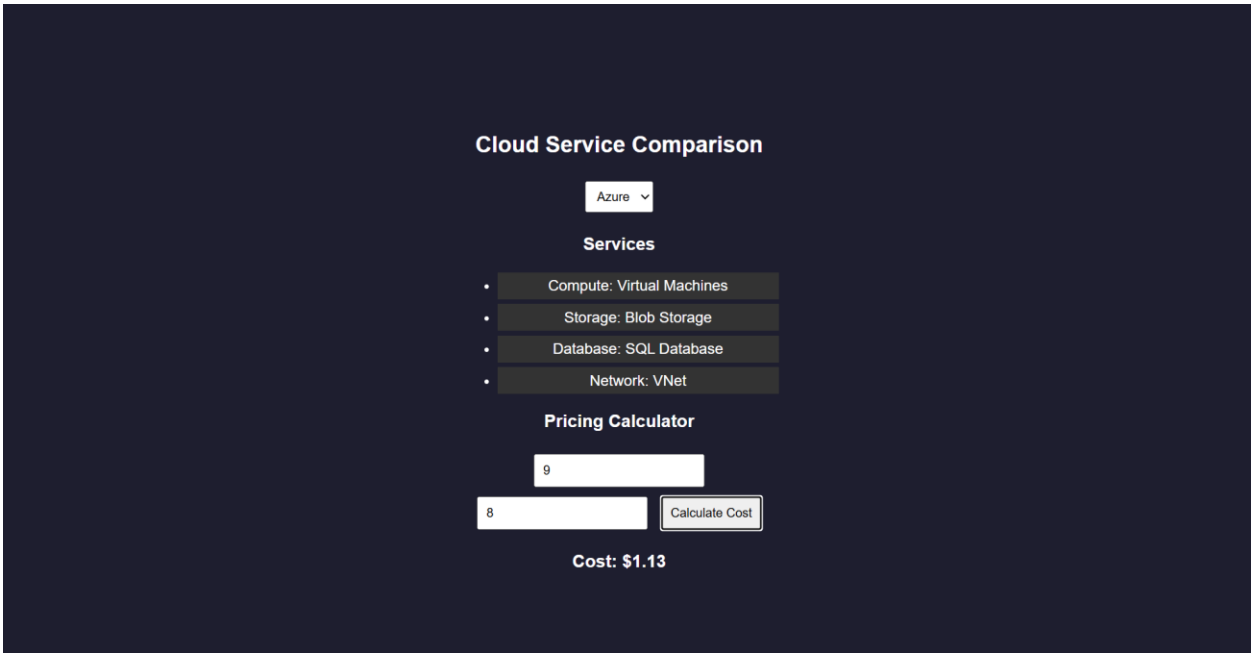
```
const services = {
    aws: {
        Compute: "EC2",
        Storage: "S3",
        Database: "RDS",
        Network: "VPC"
    },
    gcp: {
        Compute: "Compute Engine",
        Storage: "Cloud Storage",
        Database: "Cloud SQL",
        Network: "VPC"
    },
    azure: {
        Compute: "Virtual Machines",
Storage: "Blob Storage",
```

```
        Database: "SQL Database",
        Network: "VNet"
    }
};
const pricing = {
    aws: { compute: 0.1, storage: 0.02 },
    gcp: { compute: 0.09, storage: 0.025 },
    azure: { compute: 0.11, storage: 0.018 }
};
function getServices(provider) {
    return services[provider];
}
function calculate(provider, hours, storage) {
    const p = pricing[provider];
    return (hours * p.compute + storage * p.storage).toFixed(2);
}
module.exports = { getServices, calculate };
```

OUTPUT:







RESULTS: A cloud comparison application was successfully developed to display services and simulate pay-as-you-use pricing across AWS, GCP, and Azure.